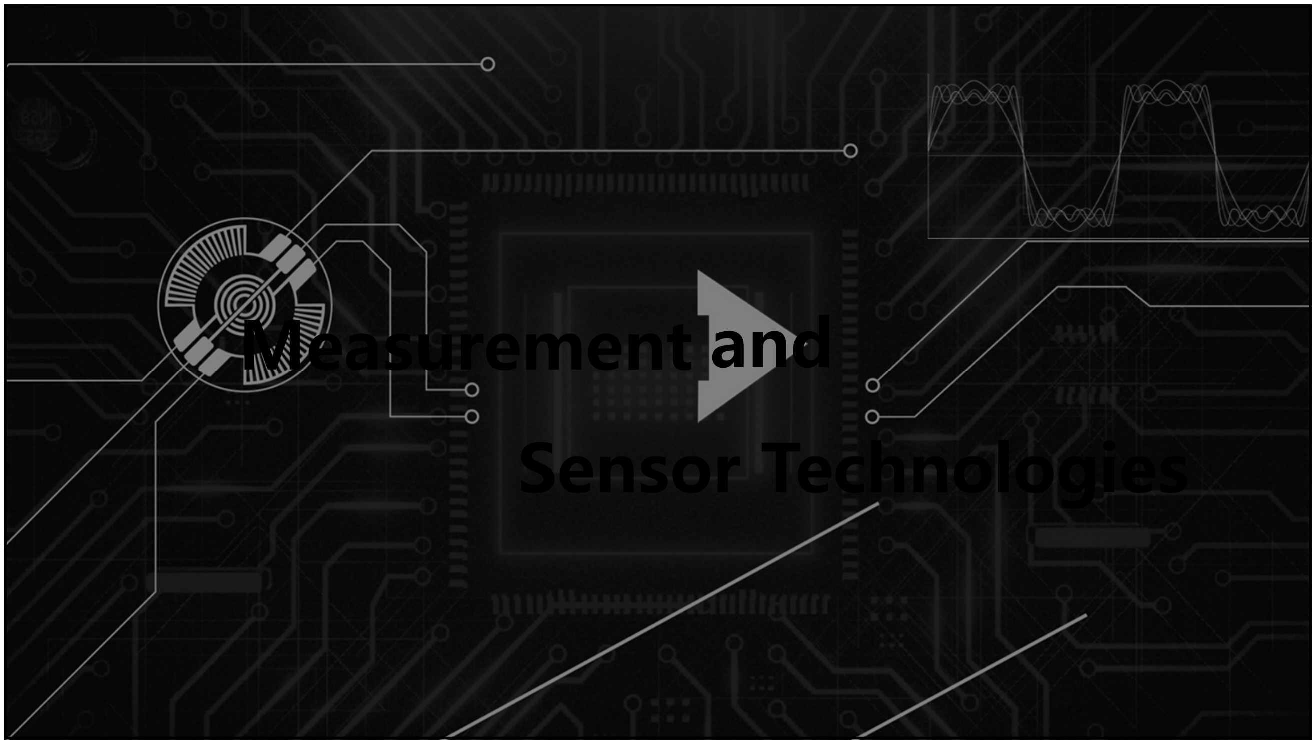


04

Basic Characteristics of Measurement Systems

Input $x(t)$ → Measurement system $h(t)$ → Output $y(t)$

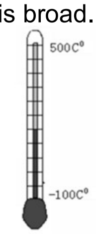
- How to describe a measurement system?
- What kind of measurement system can meet the requirements of signal analysis?

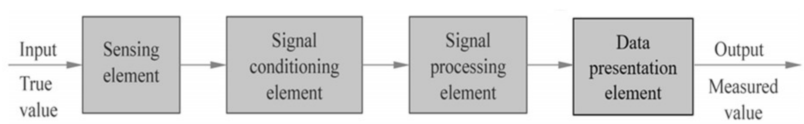
4 Basic Characteristics of Measurement Systems

4.1 Measurement System and Linear System

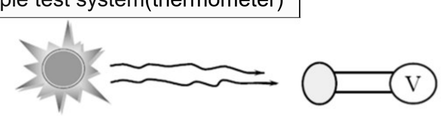
■ 4.1.1 Basic Requirements of Measurement System

The term *measurement system* in this book refers to both the complex test system consisting of many components, and the components of the measurement system, such as sensors, conditioning circuits, recording instruments, etc. Therefore, the concept of measurement system is broad.

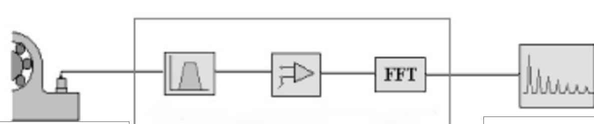




Simple test system(thermometer)



Complex test system(Bearing Defect Detection)



Bearing vibration signal

Signal spectrum

4.1 Measurement System and Linear System

■ 4.1.1 Basic Requirements of Measurement System

The output signal of the measurement system can truly reflect the change process of the measured physical quantity, without distortion of the signal, that is, to achieve **the non-distortion test（不失真测试）.中文P56**

An ideal measurement system should have a one to one, deterministic input and output relationship, that is, for each deterministic input there should be a unique output corresponding to it, and the input and output in a linear relationship is the best, moreover, the characteristics of the measurement system should not change with time, and the system meeting the above requirements is a **linear time invariant (LTI) system(线性时不变系统)**.
中文P57

Therefore, the system with linear time-invariant(LTI) characteristics is the best measurement system(最佳测试系统).

4.1 Measurement System and Linear System

■ 4.1.2 Linear Systems and Main Characteristics

- The relationship between the input x(t) and output y(t) of a linear time-invariant system can be described by a linear differential equation with constant coefficients. The general form of the differential equation is

$$a_n\frac{d^ny(t)}{dt^n}+a_{n-1}\frac{d^{n-1}y(t)}{dt^{n-1}}+\cdots+a_1\frac{dy(t)}{dt}+a_0y(t)$$

$$=b_m\frac{d^mx(t)}{dt^m}+b_{m-1}\frac{d^{m-1}x(t)}{dt^{m-1}}+\cdots+b_1\frac{dx(t)}{dt}+b_0x(t)$$

Superposition Property. $x_1(t)\to y_1(t),\quad x_2(t)\to y_2(t),$

$[x_1(t)\pm x_2(t)]\to[y_1(t)\pm y_2(t)].$

Proportional Property. $x(t)\to y(t),$

$cx(t)\to cy(t).$

Differential Property. $\frac{\mathrm{d}^n x(t)}{\mathrm{d}t^n}\to\frac{\mathrm{d}^n y(t)}{\mathrm{d}t^n}.$

Integral Property. $\int_0^t x(t)\mathrm{d}t\to\int_0^t y(t)\mathrm{d}t.$

Frequency Retention Property. The frequency retention means that the steady-state output of a linear time-invariant system has the same frequency as the input signal.

The main characteristics of linear systems, especially the superposition and frequency property, play an important role in the test work.

4.1 Measurement System and Linear System

■ 4.1.3 Transmission Characteristics of Measurement System

- Characteristics of measurement system performance are usually subdivided into two classes on the basis of the frequency of the input signals.

Basic characteristics

Static characteristics

When the input and output do not change with time or change slowly, the relationship between the output and the input can be expressed by an algebraic equation.

Dynamic characteristics

when the input and output change rapidly with time, the relationship between the output and the input can be expressed by a differential equation.

4.2 Static Characteristics of a Measurement System

■ **Static characteristics of measurement system mainly include**

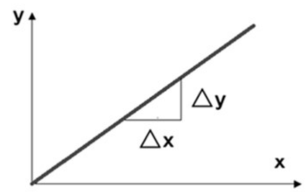
1）Sensitivit（灵敏度）
2）Linearity（线性度）
3）Hysteresis（回程误差）
4）Repeatability（重复性）
5）Accuracy（精度）
6）Drift and Stability（漂移,稳定性）
7）Resolution（分辨力/率）
8）Calibration （校准）

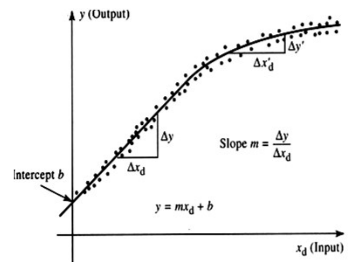
4.2 Static Characteristics of a Measurement System

4.2.1 Sensitivity

- **The sensitivity of an instrument or system is the ratio of the incremental output quantity to the incremental input quantity.**

S=△y/△x





- **The sensitivity may be constant for only part of the normal operating range of the instrument.**

4.2 Static Characteristics of a Measurement System

4.2.1 Sensitivity

- The dimension of sensitivity depends on the input-output dimension. When the dimensions of the input and output are the same, the sensitivity is a dimensionless number, often referred to as "amplification" or "gain".
- In a linear system, the sensitivity is a slope of ideal straight line.For a nonlinear system, the sensitivity is not a constant. It will vary with the input value. Usually a reference straight line is used to replace the actual characteristic curve (fitted straight line), and the slope of the fitted straight line is used as the average sensitivity of the test system.

Least squares curve fitting method (最小二乘曲线拟合)

$$y = kx + b \qquad \Delta_i = y_i - (kx_i + b)$$

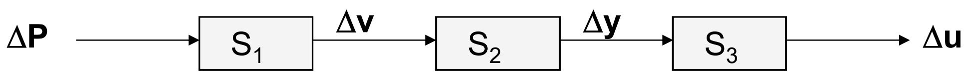
$$k = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n\Sigma x_i^{\,2} - (\Sigma x_i)^2}$$

$$b = \frac{(\Sigma x_i^{\,2}\Sigma y_i - \Sigma x_i \Sigma x_i y_i)}{n\Sigma x_i^{\,2} - (\Sigma x_i)^2}$$

4.2 Static Characteristics of a Measurement System

- 4.2.1 Sensitivity

  - Sensitivity of the multi-stage test system:



$$S=\triangle u/\triangle P=\triangle v/\triangle P\ .\ \triangle y/\triangle v\ .\ \triangle u/\triangle y=S_1\,S_2\,S_3$$

  - When the test system consists of multiple independent links, the total sensitivity is equal to the product of the sensitivity of each link.

Sensitivity reflects the ability of the test system to respond to changes in the input quantity (determined by the physical properties or structure of the test system).The higher the sensitivity, the easier it is to be affected by external interference and noise, the smaller the measurement range and the worse the stability.(reasonable choice)

4.2 Static Characteristics of a Measurement System

- 4.2.1 Sensitivity

■ Example 4.1

The following voltage values of a weight-measuring system are measured at a range of weights. Determine the measurement sensitivity of the instrument in mV/g(Table 4.1).

Table 4.1 Voltage values of a weight-measuring system

Weight (g)	10	15	20	30
Voltage (mV)	308	316	324	332

For a change in weight of 5 g, the change in voltage is 8 mV, hence the measurement sensitivity $S = 8\ \text{mV}/5\ \text{g} = 1.6\ \text{mV/g}$.

4.2 Static Characteristics of a Measurement System

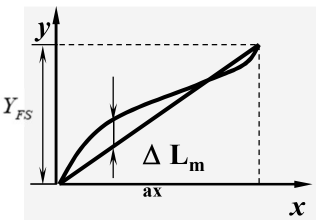
4.2.2 Linearity

- The actual test system output and input are often not ideal straight lines, such static characteristics are represented by polynomials

$$y(t) = S_0 + S_1 x + S_2 x^2 + \ldots + S_n x^n$$

- Linearity is determined by the **calibration curve(定度曲线).** The static calibration curve plots the output amplitude versus the input amplitude under static conditions. Its degree of resemblance to a straight line describes the linearity.Linearity is often specified in terms of percentage of nonlinearity, which is defined as:

$$\delta_L = \frac{|\Delta L_{max}|}{Y_{FS}} \times 100\%$$

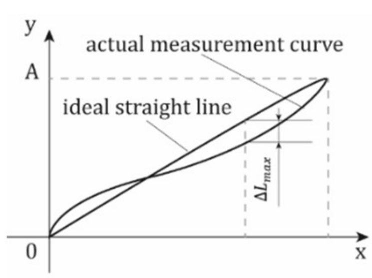


The smaller δ_L is, the better the linearity of the system!

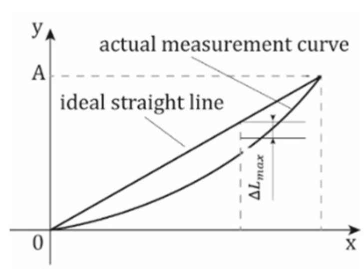
4.2 Static Characteristics of a Measurement System

4.2.2 Linearity

- There are different methods to obtain the ideal curve. One way is to use terminal points method(端基直线), that is, to determine output values at the smallest and highest input values and draw a straight line through these two points. Another is to find the straight line by using the method of least squares to determine the best fit one(最小二乘拟合直线) when all data values are considered equally likely to be in error.



Terminal Points Method



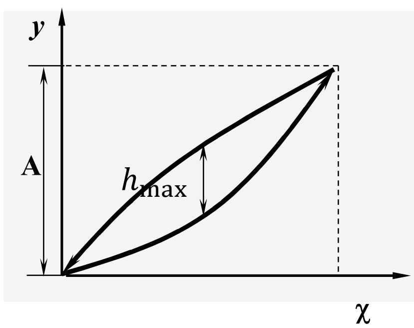
Least Squares Method

4.2 Static Characteristics of a Measurement System

4.2.3 Hysteresis(回程误差)

- Instruments can give different outputs from the same value of quantity being measured according to whether that value has been reached by a continuously increasing change or a continuously decreasing change. This effect is called hysteresis.Figure shows such an output with the hysteresis as the maximum difference in output for increasing and decreasing values.

$$\delta_{\mathrm{H}} = \frac{|h_{\max}|}{A} \times 100\%$$

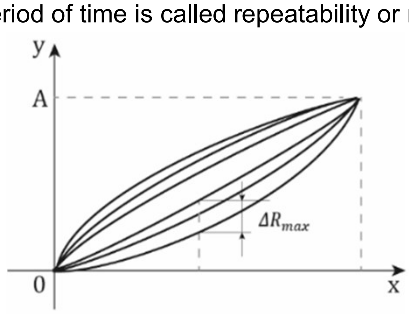


- Describes the characteristics of the output of the system in relation to the direction in which the input changes.
- Reasons for hysteresis: magnetic hysteresis, elastic hysteresis, gap, friction, material deformation, etc.
- 原因：中文P61

4.2 Static Characteristics of a Measurement System

4.2.4 Repeatability(重复性)

- The ability of an instrument to give the same output for equal inputs applied over some period of time is called repeatability or reproducibility.



$$\delta_{\mathrm{R}} = \frac{|\Delta R_{\max}|}{A} \times 100\%$$

- Repeatability is often directly related to accuracy. But a sensor can be inaccurate, yet be repeatable in making observations.

- A measure of the dispersion of measurement results, that is, an indicator of the size of random errors. It can also be expressed as:

$$\delta_R = \frac{(2 \sim 3)\sigma}{A} \times 100\%$$

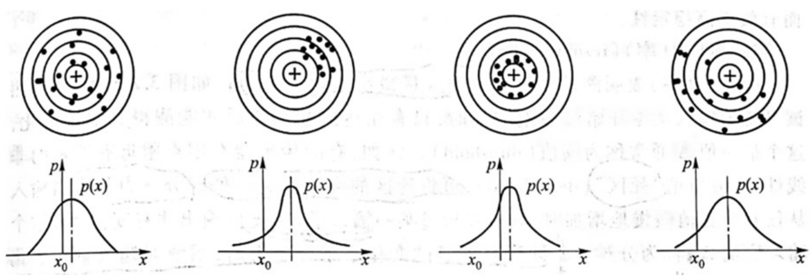
σ——Standard deviation of measured values

4.2 Static Characteristics of a Measurement System

- 4.2.5 Accuracy（精度）

Accuracy represents the degree of agreement between the measurement results of the test system and the true value and reflects the comprehensive influence of systematic errors and random errors. （中文P61）

Accuracy(精确度)**=correctness(**准确度**)+precision(**精密度**)**



4.2 Static Characteristics of a Measurement System

- 4.2.5 Accuracy

- The accuracy of a single measured quantity is the difference between the true value and the measured value divided by the true value. The index for evaluating the size of the measurement error generated by the test system is mainly characterized by the measurement error(测量误差).
- ◆ The smaller the measurement error, the higher the measurement accuracy.
- In engineering, the reference error（引用误差） is often used as the scale for judging the accuracy level. The allowable reference error (maximum reference error: the ratio of the maximum absolute error to the full scale) is used as the code for the accuracy level.

For example, a 0.2-level voltmeter means that the allowable indication error of the voltmeter does not exceed ±0.2% of the range of the voltmeter.

$$|\gamma_m| \le G\%$$

4.2 Static Characteristics of a Measurement System

- 4.2.5 Accuracy（中文P62）

  - Reference Error: Absolute error as a percentage of the ratio of full scale

$$\delta_n = \frac{|\Delta|}{F.S.} \times 100\%$$

Maximum Reference error: $$\delta_{nm} = \frac{|\Delta_m|}{A} \times 100\ \%$$

- Accuracy grade G: Indicates that the test system or device can keep its error within the specified limit when it meets certain metrological requirements.

- The accuracy grade represents the size of the allowable error, not the error that occurs in the actual measurement.

$$\delta_{nm} \le \alpha\ \% \qquad |\gamma_m| \le G\%$$

4.2 Static Characteristics of a Measurement System

- 4.2.5 Accuracy

  - [Example 1] The accuracy level of a pressure gauge is 1.5, the range is 0 ~ 2.0MPa, and the measurement result is displayed as 1.2MPa, find 1) the maximum reference error δ_{nm}; 2) the maximum absolute error that may occur Δ_{m}; 3) the Indication relative error δ_x =?

[Solution] 1) The maximum reference error can be directly obtained from the accuracy level, that is,

2)

3)

4.2 Static Characteristics of a Measurement System

4.2.5 Accuracy

- [Example 1] The accuracy level of a pressure gauge is 1.5, the range is 0 ~ 2.0MPa, and the measurement result is displayed as 1.2MPa, find 1) the maximum reference error δ_{nm}; 2) the maximum absolute error that may occur Δ_{m}; 3) the Indication relative error δ_x =?

[Solution] 1) The maximum reference error can be directly obtained from the accuracy level, that is,

$$\delta_{\mathrm{nm}} = 1.5\%$$

2) $$\Delta_{\mathrm{m}} = \pm 2 \times 1.5\% = \pm 0.03\mathrm{MPa}$$

3) $$\delta_x = \frac{|\Delta_m|}{x} \times 100\% = \frac{0.03}{1.2} \times 100\% = 2.5\%$$

4.2 Static Characteristics of a Measurement System

4.2.6 Drift and Stability（漂移和稳定性 中文P62）

- Drift is the deviation from a specific reading of the sensor when the sensor is kept at that value for a prolonged the period of time.
- Stability is another way of stating drift. That is, with a given input, you always get the same output. Stability is the ability of a sensor device to give same output when used to measure a constant input over a period of time.

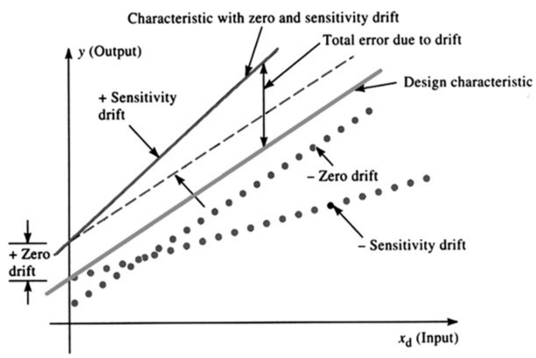
Stability (internal components): Usually expressed as the ratio of the change in the output indication to time.
For example: 1.3mV/8h means that the voltage fluctuates 1.3mV every 8 hours.
Drift (external environment and working conditions): Usually expressed as the change of indication value under corresponding conditions.
For example: 0.02% F.S.(/h·°C) means that the temperature changes by 1°C, and the sensor output changes to 0.02% of the full scale value per hour.

- Reasons for drift may be different, such as changes in ambient pressure, humidity, temperature etc., or due to changes in the constituents of the sensor itself, such as aging, wear etc.

4.2 Static Characteristics of a Measurement System

4.2.6 Drift and Stability

- Zero drift describes the effect where the zero reading of the instrument in this modified by a change in ambient conditions. Zero drift refers to the change in sensor output. If the input is kept steady at a level that yields a zero reading.

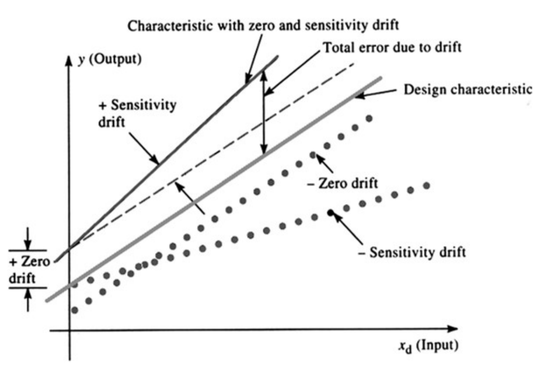


- Zero drift has occurred when all output values increase or decrease by the same absolute amount.
- The slope of the sensitivity curve is unchanged, but the output-axis intercept increases or de-creases.
- The following factors can cause zero drift: manufacturing misalignment（未对准）, variations in ambient temperature（环境温度）, hysteresis（迟滞）, vibration, shock, and sensitivity to forces from undesired directions.

4.2 Static Characteristics of a Measurement System

4.2.6 Drift and Stability

- The most common drift is temperature drift, that is, the change of output caused by the change of surrounding temperature, which further causes the sensitivity of the system to drift, that is, sensitivity drift.



- When the slope of the calibration curve changes as a result of an interfering and/or modifying input, a drift in sensitivity results.
- Sensitivity drift causes error that is proportional to the magnitude of the input. The slope of the calibration curve can either increase or decrease
- Sensitivity drift can result from manufacturing tolerances, variations in power supply, nonlinearities, and changes in ambient temperature and pressure.

4.2 Static Characteristics of a Measurement System

- 4.2.7 Resolution(分辨率)　中文P63

- The smallest incremental quantity that can be measured with certainty is the resolution.
- Resolution expresses the degree to which nearly equal values of a quantity can be discriminated.

  the smallest change in a measured variable to which an instrument will respond.
- For digital instruments, when the input quantity changes continuously, the output quantity changes in steps. Generally, it can be considered that the value represented by the last digit of the output display scale is its resolution. For example, the temperature display of a digital thermometer is 180.6°C. , the resolution is 0.1°C;
- For analog instrument, that is, a device with a continuously changing output, the resolution refers to the smallest input increment that the test device can display or record, generally half of the smallest division value(最小分度值的一半).

4.2 Static Characteristics of a Measurement System

- 4.2.8 Calibration(校准)

- Under the specified standard working conditions, the process of obtaining the static characteristic curve of the test system by the experimental method.**在规定的标准工作条件下，用实验方法求测试系统的静态特性曲线的过程。**
- When there are some random factors in the test device, repeated measurements can be made under the same conditions, and the average value under the same input conditions can be obtained to make a static characteristic curve.测试装置本身存在某些随机因素时，可在相同条件下进行多次重复测量，求同一输入条件下的平均值，作静态特性曲线。
- For a test device with hysteresis, the forward stroke and the reverse stroke form a cycle. Under the same conditions, multiple cycles are measured, and the average value is obtained to obtain the static characteristic curve of the forward and reverse strokes.有回差的测试装置，正行程和反行程组成一个循环。相同条件下多次循环测量，求出平均值，得到正反行程的静态特性曲线。

4.2 Static Characteristics of a Measurement System

■ 4.2.8 Calibration

- The procedures of calibration and recalibration are the same as following:

(1) Examine the construction of the measurement system, and identify and list all the possible inputs.

(2) Choose, as best as you can, which of the inputs will be important in the application for which the measurement system is to be calibrated.

(3) Obtain apparatus that will allow you to vary all the significant inputs over ranges considered necessary. Procure standards to measure each input.

(4) By holding some inputs constant, varying others, and recording the output(s), develop the desired static input-output relations.